# Chapter 9

# Wide Area Networks

# Outline

9.1 - Introduction

9.2 - Dedicated-Circuit Networks

9.3 - Packet-Switched Networks

9.4 - Virtual Private Networks

9.5 - Best practice WAN design

9.6 - Improving WAN Performance

9.7 - Implications for Cyber Security

# 9.1 Introduction

- Wide area networks (WANs)
  - Connect BNs and LANs across longer distances, often hundreds of miles or more
- Typically built by using leased circuits from common carriers such as AIS, UIH
  - Most organizations cannot afford to build their own WANs

# Introduction (Cont.)

- Focus of the Chapter
  - Examine WAN architectures and technologies from a network manager point of view
- Regulation of services
  - Federal Communications Commission (FCC) in the US
  - Canadian Radio Television and Telecomm Commission (CRTC) in Canada
  - Public Utilities Commission (PUC) in each state
- Common Carriers
  - Local Exchange Carriers (LECs)
  - Interexchange Carriers (IXCs)

# Services Used by WANs

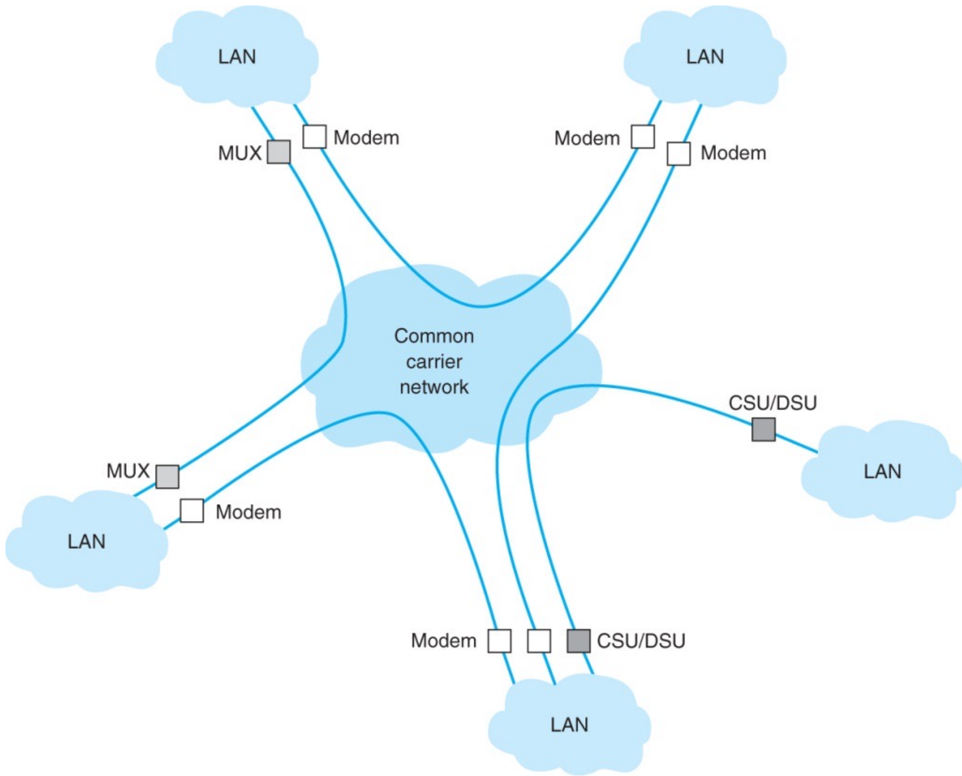
- Use common carrier networks
  – Dedicated-Circuit Networks
  – Packet-Switched Networks
- Use public networks
  – Virtual Private Networks

# 9.2 Dedicated Circuits

- Leased full duplex circuits from common carriers
- Used to create point to point links between organizational locations
  - Routers and switches used to connect these locations together to form a network
- Billed at a flat fee per month (with unlimited use of the circuit)
- Require more care in network design
- Basic dedicated circuit architectures
  - Ring, star, and mesh
- Dedicated Circuit Services
  - T carrier services
  - Synchronous Optical Network (SONET) services

# Dedicated Circuit Services




Equipment installed at the end of dedicated circuits

- CSU/DSU: Channel Service Unit / Data Service Unit
- WAN equivalent of a NIC in a LAN
- May also include multiplexers

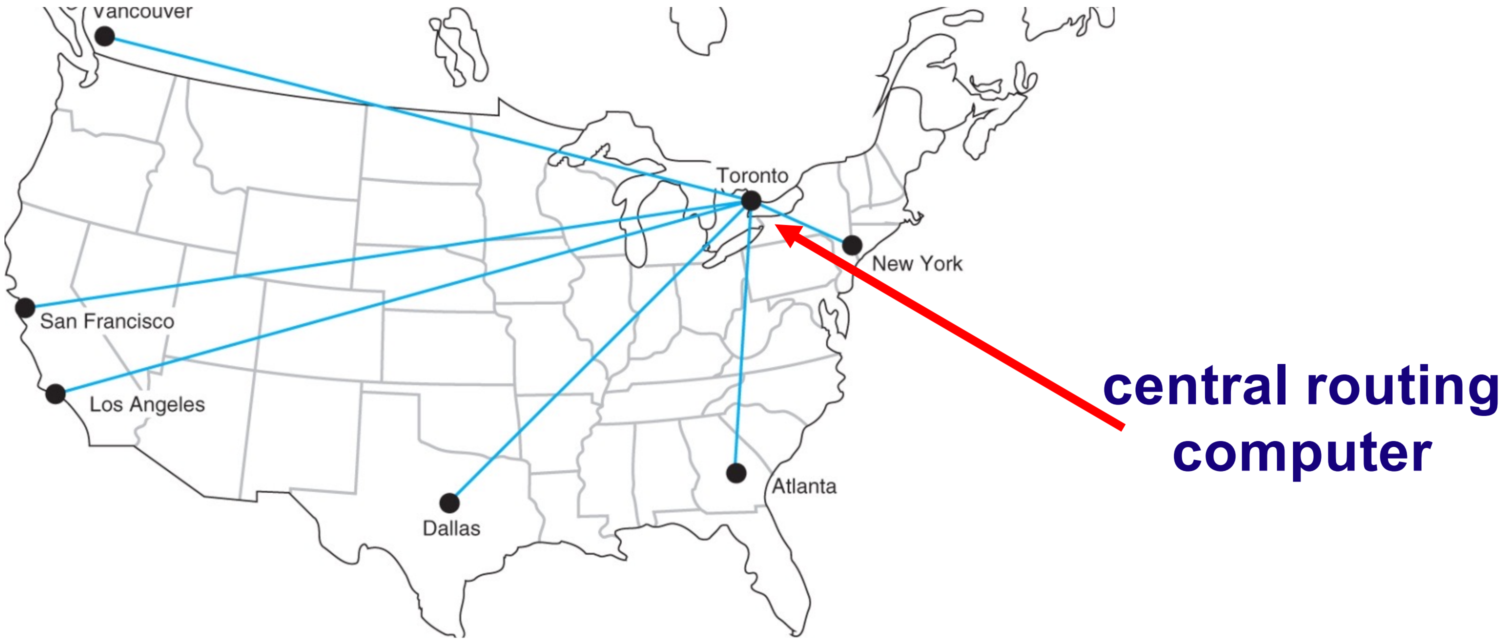
# Ring Architecture

- Reliability
  - Data can flow in both directions (full-duplex circuits)
    - With the expense of dramatically reduced performance
- Performance
  - Messages travel through many nodes before reaching destination

# Star Architecture

- Easy to manage
  - Central computer routes all messages in the network
- Reliability
  - Failure of central computer brings the network down
  - Failure of any circuit or computer affects one site only
- Performance
  - Central computer becomes a bottleneck under high traffic

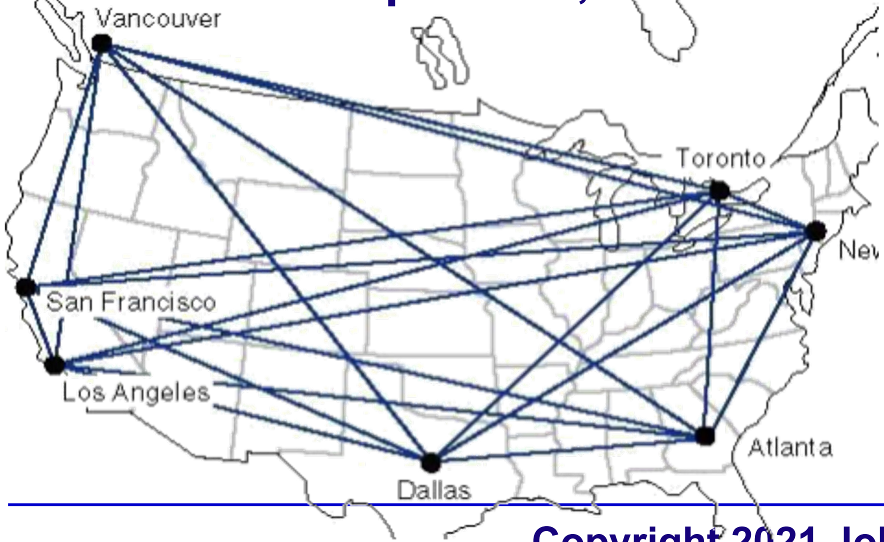


# Mesh Architectures

- Combine performance benefits of ring and star networks
- Use decentralized routing, with each computer performing its own routing
- Impact of losing a circuit is minimal (because of the alternate routes)
- More expensive than setting up a star or ring network.

**Full mesh**

- Expensive, seldom used




**Partial mesh**

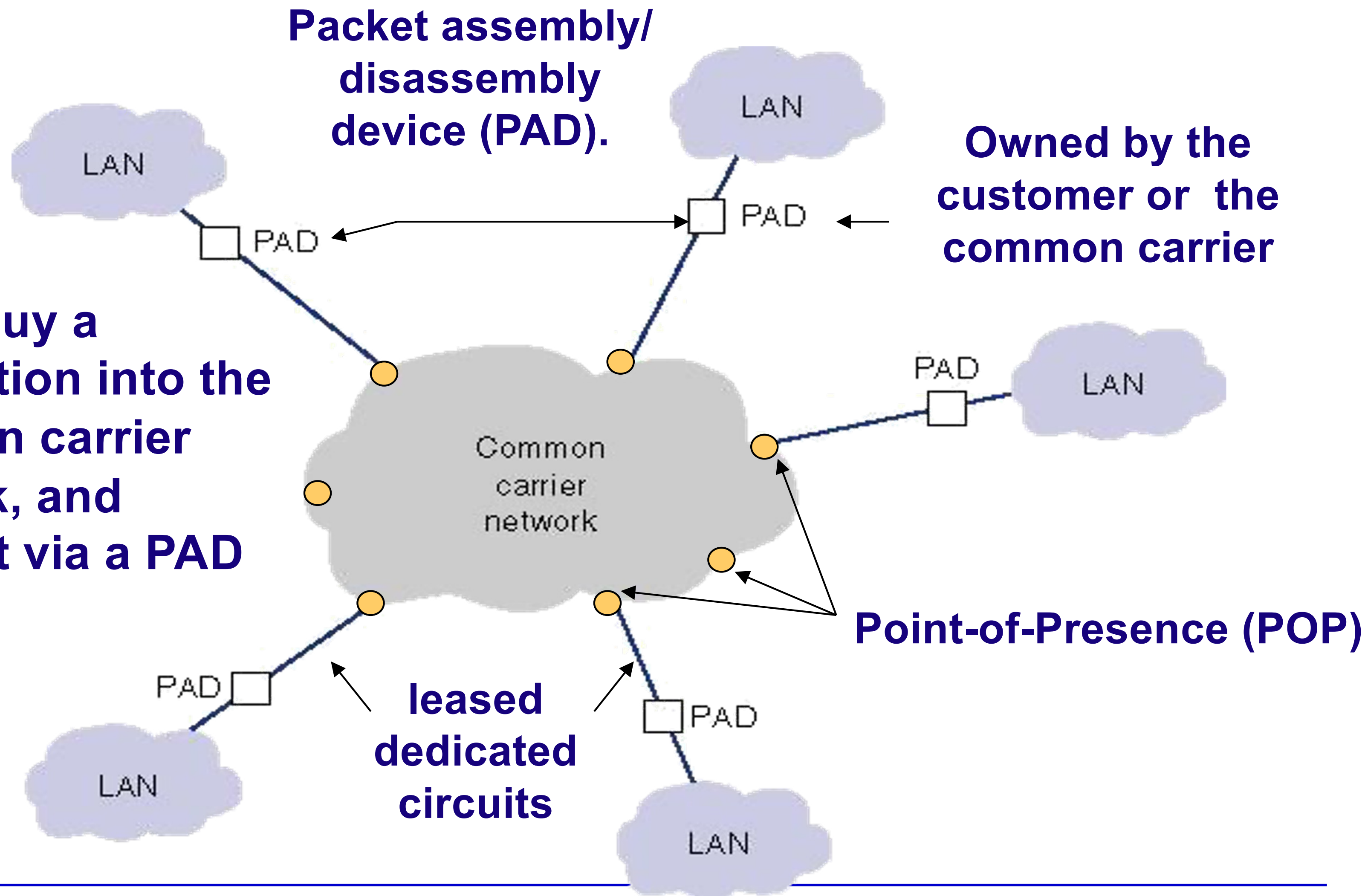
- More practical

# 9.3 Packet Switched Services

- In both circuit switched and dedicated services
  - A circuit is established between two computers
    - Solely dedicated or assigned for use only between these two computers
- Packet switched services
    - Enable multiple connections to exist simultaneously between computers over the same physical circuits
    - User pays a fixed fee for the connection to the network plus charges for packets transmitted

# Basic Architecture of Packet Switched Services

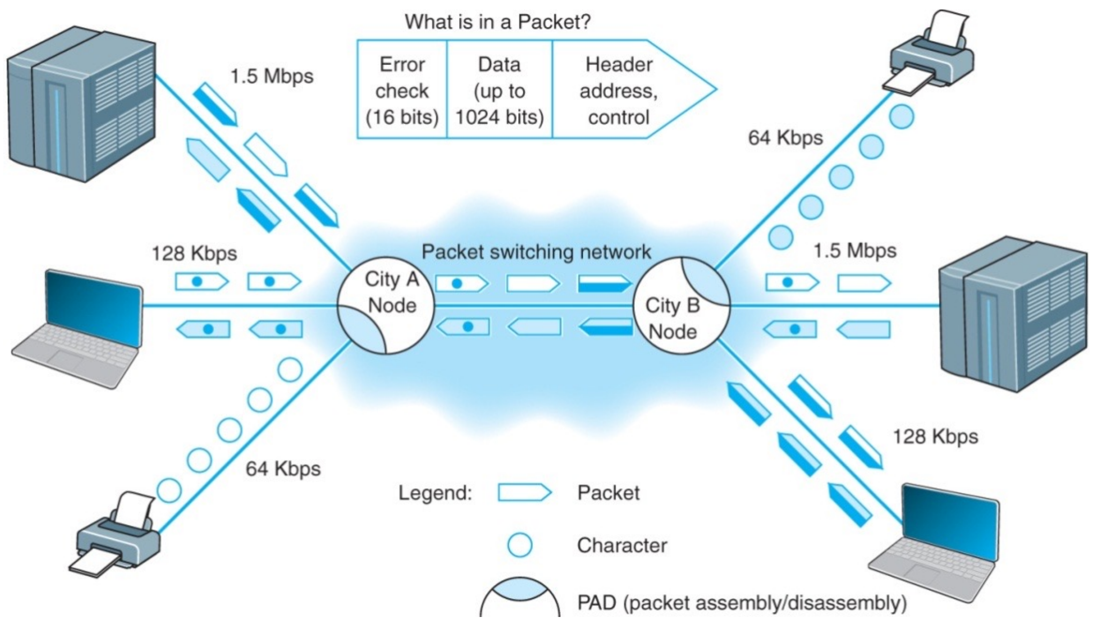
Users buy a connection into the common carrier network, and connect via a PAD



# Packet Switching

- Interleave packets from separate messages for transmission
  - Most data communications consists of short burst of data
  - Packet switching takes advantage of this burstiness
    - Interleaving bursts from many users to maximize the use of the shared network



# Packet Routing Methods

- Describe which intermediate devices the data is routed through
- Connectionless (Datagram)
  - Adds a destination and sequence number to each packet
  - Individual packets can follow different routes through the network
  - Packets reassembled at destination
- Connection Oriented (Virtual Circuit (VC))
  - Establishes an end-to-end circuit between the sender and receiver
  - All packets for that transmission take the same route over the virtual circuit established
  - Same physical circuit can carry many VCs

# Types of Virtual Circuits

- Permanent Virtual Circuit (PVCs)
  - Established for long duration (days or weeks)
  - Changed only by the network manager
  - More commonly used
  - Packet switched networks using PVCs behave like a dedicated circuit networks
- Switched Virtual Circuit (SVC)
  - Established dynamically on a per call basis
  - Disconnected when the call ends

# Data Rates of Virtual Circuits

- Users specify the rates per PVC via negotiations
  - Committed information rate (CIR)
    - Guaranteed by the service provider
    - Packets sent at rates exceeding the CIR are marked discard eligible (DE)
      - discarded if the network becomes overloaded
  - Maximum allowable rate (MAR)
    - Sends data only when the extra capacity is available

# Packet Switched Service Protocols

- Frame Relay
- IP/MPLS
- Ethernet Services

- Several common carriers announced they will stop offering all but Ethernet and Internet services soon

# Frame Relay

- One of the oldest used packet services. Most companies are no longer installing it. It is legacy technology.
- Uses T-carrier and SONET (1.5 Mbps, 45 Mbps, 155 Mbps, 622 Mbps)
- Unreliable packet service because it does not perform error control. It checks errors but simply discard. It is up to the end-points to control the errors.
- Committed Information Rate, CIR, and Maximum Allowable Rate, MAR
  - Packet that exceed CIR will be marked as Discard Eligible, DE.
  - If the network becomes overloaded DE packet will be discard.

# Multi Protocol Label Switching (MPLS)

- Does not fit to 5-layer model.
- designed to work with a variety of commonly used layer 2 protocols.
- Commonly used with IP for routing.
  - Strip off Ethernet frame and use IP address to route the packet through the carrier's packet-switched network
- Uses either T-carrier or SONET as its wiring (1.5 Mbps, 45 Mbps, 155 Mbps, 622 Mbps)

# MPLS – How It Works

- The customer connects to the common carrier's network using any common layer 2 service
  - (e.g., T carrier, SONET, ATM, frame relay, Ethernet)
- The carrier's switch at the network entry point examines the incoming frame and converts the incoming layer 2 or layer 3 address into an MPLS address label
- The carrier can use the same layer 2 protocol inside its network as the customer, or it can use something different
- When delivered, the MPLS switch removes the MPLS header and delivers the packet into the customer's network using whatever layer 2 protocol the customer has used to connect into the carrier's network at this point (e.g., frame, T1).

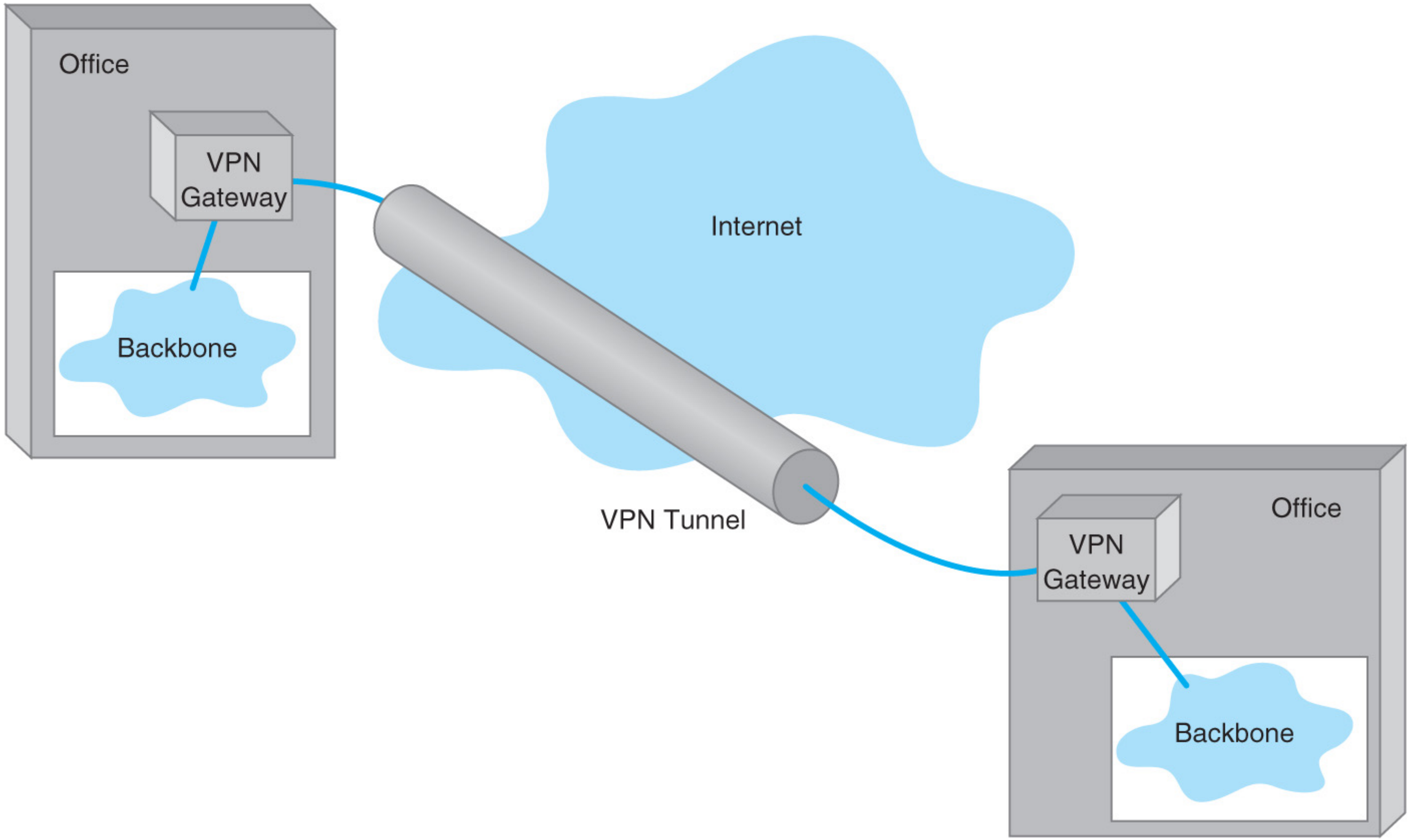
# Ethernet Services

- Most organizations use Ethernet and IP in the LAN and BN.
- Ethernet Services differ from WAN packet services like Frame Relay or MPLS
  – No need to translate LAN protocol (Ethernet/IP) to the protocol used in WAN services
  – Frame Relay use different protocols requiring translation from/to LAN protocols
- Currently offer CIR speeds from 1 to 10 Gbps at a lower cost than traditional services
- Emerging technology; expect changes

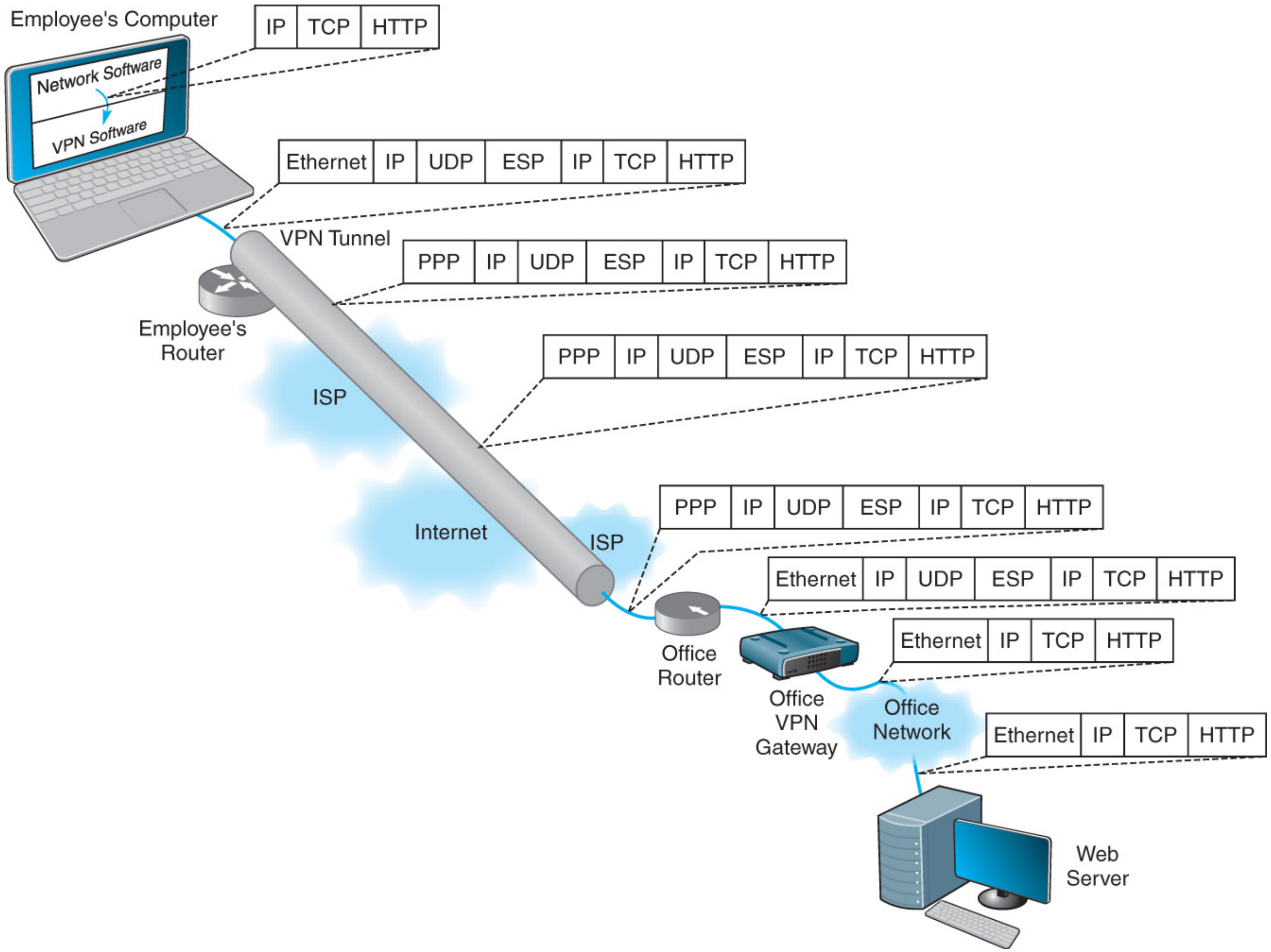
# 9.4 Virtual Private Networks

- Provides equivalent of a private packet switched network over public Internet
  - Use Permanent Virtual Circuits (*tunnels*) that run over the public Internet, yet appear to the user as private networks
  - Encapsulate the packets sent over these tunnels using special protocols that also encrypt the IP packets
- Provides low cost and flexibility
- Disadvantages of VPNs:
  - Unpredictability of Internet traffic
  - Lack of standards for Internet-based VPNs, so that not all vendor equipment and services are compatible

# VPN Architecture



# VPN Encapsulation of Packets



# VPN Types

- Intranet VPN
  - Provides virtual circuits between organization offices over the Internet
- Extranet VPN
  - Same as an intranet VPN except that the VPN connects several different organizations, e.g., customers and suppliers
- Access VPN
  - Enables employees to access an organization's networks from remote locations

# 9.5 WAN Design Practices

- Difficult to recommend best practices
  - Services, not products, being bought
  - Fast changing environment with introduction of new technologies and services from non-traditional companies

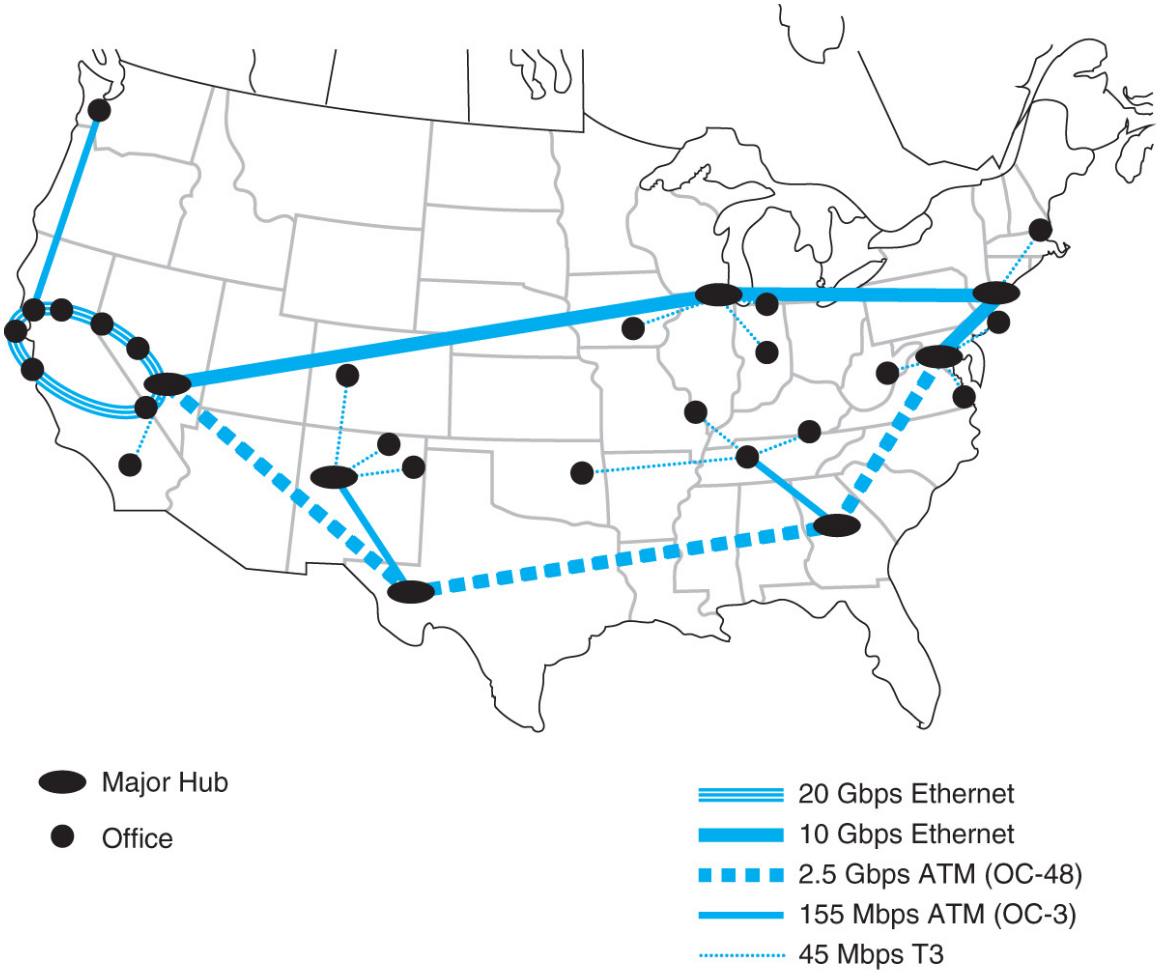
# Software Defined WAN, SDWAN

- Used software to and manage the routers used in WAN.
  - Balance the use of different network services
  - Software trade-offs between  higher cost (e.g. packet switched services vs the less reliable Internet-based VPN that are cheaper.

# SDWAN Benefits

1) Centralized management software system to see all network status from all devices in WAN.
    - Manager can define rules for network decision.
2) Reducing cost by balancing network traffic over different WAN services.
3) See end-to-end traffic flows to better predict future growth.
4) Good security because most routers that are compatible to SDWAN have built-in VPN capabilities of encryption from router-to-router in SDWAN

# WAN Services



# WAN Services

| Type of Services | Data Rates | Relative Cost | Reliability |
|---|---|---|---|
| Dedicated-Circuit Services | | | |
| T-Carrier | 64 kbps to 45 Mbps | Moderate | Reliable |
| SONET | 50 Mbps to 10 Gbps | Moderate | Reliable |
| Packet-Switched Services | | | |
| Ethernet | 1 Mbps to 40 Gbps | Moderate | Reliable |
| MPLS | 64 kbps to 1 Gbps | Moderate | Reliable |
| VPN services | | | |
| VPN | 3 Mbps to 300 Mbps | Low | Not Reliable |

# Recommendations for the Best WAN Practices

| Network Needs | Recommendation |
|---|---|
| Low to Typical Traffic (10 Mbps or less) | VPN if reliability is less important. Ethernet or MPLS otherwise |
| High Traffic (10 – 50 Mbps) | Ethernet or MPLS T3 if traffic is stable or predictable |
| Very High Traffic (more than 50 Mbps) | Ethernet or MPLS SONET if traffic is stable or predictable |

# 9.6 Improving WAN Performance

- Handled in the same way as improving LAN performance
  - Improve device performance
  - Improve circuit capacity
  - Reduce network demand

# Improving Device Performance

- Upgrade the devices (routers) and computers that connect backbones to the WAN
  - Select devices with lower "latency"
    - Time it takes in converting input packets to output packets
- Examine the routing protocol (static or dynamic)
  - Dynamic routing
    - Increases performance in networks with many possible routes from one computer to another
    - Better suited for "bursty" traffic
    - Imposes an overhead cost (additional traffic)
      - Reduces overall network capacity
      - Should not exceed 20%

# Improving Circuit Capacity

- Analyze the traffic to find the circuits approaching capacity
  - Upgrade overused circuits
  - Downgrade underused circuits to save cost
- Examine why circuits are overused
  - Caused by traffic between certain locations
    - Add additional circuits between these locations
  - Capacity okay generally, but not meeting peak demand
    - Add a circuit switched or packet switched service that is only used when demand exceeds capacity
  - Caused by a faulty circuit somewhere in the network
    - Replace and/or repair the circuit
- Make sure that circuits are operating properly

# Reducing Network Demand

- Determine impact on network
  - Require a network impact statement for all new application software
- Use data compression of all data in the network
- Shift network usage
  - From peak or high cost times to lower demand or lower cost times
  - e.g., transmit reports from retail stores to headquarters after the stores close
- Redesign the network
  - Move data closer to applications and people who use them
  - Use distributed databases to spread traffic across

# 9.7 Implications for Cyber Security

- WAN is one of the most secure parts of network from the common carriers.
- The cost encryption is low. VPN over the Internet or on WAN must be encrypted.